Notes: DellaVigna and Kaplan (QJE, 2007)

The Fox News Effect: Media Bias and Voting

Contents

1 What the paper does **3**

2 Notation (what the symbols mean) **3**

3 All models (equations) in the paper **4**

3.1 (1) Selection into Fox News carriage (linear probability model) 4

3.2 (2) Main DiD for presidential vote share 4

3.3 (3) Turnout regression 5

3.4 (4) Senate vote share regression (test for general ideology vs candidate-specific effects) 5

3.5 (5)–(6) Persuasion-rate model 5

3.6 (7) "First stage" for Fox News audience (Scarborough) 6

3.7 (8) Generic persuasion formula used for turnout field experiments 6

4 Figures and tables: what each one is saying (my interpretation) **6**

4.1 Figure I: Fox News availability by county in 2000 6

4.2 Table I: Summary statistics on Fox News audience (Scarborough data) 7

4.3 Table II: Summary statistics on cable and election data (town level) 8

4.4 Table III: Determinants of Fox News availability (selection test) 9

4.5 Table IV: Main result — Fox News and 2000–1996 presidential vote-share change . . 9

4.6 Table V: Robustness and persistence 10

4.7 Table VI: Interactions (heterogeneity) and placebo tests 10

4.8 Table VII: Turnout and Senate elections 11

4.9 Table VIII: Effect of Fox carriage on Fox audience + implied persuasion rates 12

4.10 Table IX: Comparing persuasion rates in other studies 13

4.11 Appendix II: Full list of controls and coefficients (for Tables III and IV benchmark specs) . . . 13

4.12 Appendix III: Why earlier drafts found smaller/null effects . . . 15

5 Takeaway **15**

6 Conclusion and summary **16**

6.1 What the paper concludes (in plain language) . . . 16

6.2 Final summary of the paper (what it does + what it finds) . . . 16

6.3 Intuition (why their design is supposed to work) . . . 17

6.4 What mechanism is most consistent with the evidence? . . . 18

6.5 How the authors interpret the persuasion effect (rational vs. behavioral) . . . 18

6.6 What the paper leaves open . . . 18

1 What the paper does

- Goal: test whether *media bias* affects voting, using the staggered entry of the conservative Fox News Channel into local cable systems between 1996 and 2000.
- Unit of analysis: U.S. *town* (9,256 towns across 28 states), matched to cable-lineup data (Nov. 2000), election data (esp. 1996 and 2000), and Census demographics (1990/2000).
- Identification idea: conditional on geography and cable-system size, which *towns* got Fox News by 2000 is argued to be "as good as idiosyncratic" within districts/counties.
- Main finding: Fox News availability in 2000 increases the Republican two-party presidential vote share change from 1996 to 2000 by about 0.4–0.7 percentage points (depending on fixed effects).
- Extra outcomes: turnout (mixed but some evidence of increases) and Senate Republican vote share (positive, suggesting an ideological shift beyond just candidate-specific coverage).
- They translate the voting effect into persuasion rates using audience data (Scarborough). Depending on audience measure, they get persuasion rates in the ballpark of 3–8% (recall audience) or 11–28% (diary audience).

2 Notation (what the symbols mean)

Town index: k.

Years: $t \in \{1996, 2000\}$ unless otherwise stated.

Key treatment:

$$d_{k,2000}^{FOX} = \begin{cases} 1 & \text{if Fox News is in the local cable package in town } k \text{ by 2000,} \\ 0 & \text{otherwise.} \end{cases}$$

Presidential vote share:

$$v_{k,t}^{R,Pres} = \frac{\text{Rep votes in presidential election}}{\text{Rep votes + Dem votes}} \quad \text{(two-party share).}$$

Turnout measures used in different places:

- In selection regressions: "log turnout" is log(total votes / voting-age population).
- In turnout regressions: $t_{k,t}^{Pres} = \ln(V_{k,t}^{TOT,Pres})$ is log total votes cast; they control for population changes separately.

Controls:

- $X_{k,2000}$: Census demographics in 2000 (12 variables).
- $X_{k,00-90}$: changes from 1990 to 2000 in those demographics.

- $C_{k,2000}$: cable-system controls (deciles of # channels and potential subscribers).

Estimation conventions:

- Regressions are typically weighted by votes cast in the 1996 presidential election.
- Standard errors are clustered by local cable company.
- Geographic fixed effects: either U.S. House district FE or county FE in key specs.

3 All models (equations) in the paper

3.1 (1) Selection into Fox News carriage (linear probability model)

$$d_{k,2000}^{FOX} = \alpha + \beta\, v_{k,1996}^{R,Pres} + \gamma\, t_{k,1996}^{Pres} + \Gamma_{2000} X_{k,2000} + \Gamma_{00-90} X_{k,00-90} + \Gamma_C C_{k,2000} + \varepsilon_k. \tag{1}$$

Interpret (1):

- This is a *selection test*: do pre-Fox political variables (1996 vote share and turnout) predict Fox availability in 2000?
- If β or γ remain large after controls and geography FE, that would suggest endogenous rollout (Fox entering where Republicans were already trending up).
- The paper's key claim: once I condition on cable size + geography (district/county FE), β and γ are basically 0 $\Rightarrow$ within-area rollout looks idiosyncratic.

3.2 (2) Main DiD for presidential vote share

$$\Delta v_k^{R,Pres} \equiv v_{k,2000}^{R,Pres} - v_{k,1996}^{R,Pres} = \alpha + \delta_F\, d_{k,2000}^{FOX} + \Gamma_{2000} X_{k,2000} + \Gamma_{00-90} X_{k,00-90} + \Gamma_C C_{k,2000} + \varepsilon_k. \tag{2}$$

Interpret (2):

- δ_F is the *Fox News effect* on the change in Republican presidential vote share from 1996 to 2000.
- With district FE or county FE, identification is basically within those geographic units, comparing towns that do vs. do not have Fox by 2000.
- A positive δ_F means Fox News availability shifted votes toward Republicans (either via persuasion, turnout changes, or both).

3.3 (3) Turnout regression

$$\Delta t_k^{Pres} \equiv t_{k,2000}^{Pres} - t_{k,1996}^{Pres} = \alpha + \delta_F\, d_{k,2000}^{FOX} + \beta\,[\ln(Pop_{k,2000}) - \ln(Pop_{k,1996})] \\ + \Gamma_{2000} X_{k,2000} + \Gamma_{00-90} X_{k,00-90} + \Gamma_C C_{k,2000} + \varepsilon_k, \tag{3}$$

where $t_{k,t}^{Pres} = \ln(V_{k,t}^{TOT,Pres})$.

Interpret (3):

- Outcome is percent change in *total votes cast* (in logs), net of population growth.
- $\delta_F > 0$ here means Fox News *increases turnout*.

3.4 (4) Senate vote share regression (test for general ideology vs candidate-specific effects)

$$v_{k,2000}^{R,Sen} = \alpha + \rho\, v_{k,1996}^{R,Pres} + \delta_F\, d_{k,2000}^{FOX} + \theta_F\, \left(d_{k,2000}^{FOX} \times d_{NY}\right) \\ + \Gamma_{2000} X_{k,2000} + \Gamma_{00-90} X_{k,00-90} + \Gamma_C C_{k,2000} + \varepsilon_k. \tag{4}$$

Interpret (4):

- Senate outcomes are used because most Senate races are not heavily covered nationally.
- δ_F is the effect on *non-featured* Senate races (ideological channel).
- θ_F captures any *extra* effect in the featured New York Senate race (Clinton–Lazio), which Fox covered a lot.
- Finding $\theta_F \approx 0$ suggests Fox shifts general ideology rather than only changing beliefs about the specific candidate(s) it covers intensely.

3.5 (5)–(6) Persuasion-rate model

Setup (their stylized accounting):

- Before Fox: fraction Republican voters r, Democratic voters d, nonvoters $1 - r - d$.
- Exposure to Fox News in town $j \in \{T, C\}$ is e_j (higher in treatment towns).
- Persuasion rate f: among *non-Republicans* who are exposed, fraction f are convinced to vote Republican.

Two-party Republican vote share in town j:

$$v_j = \frac{r + (1-r)e_j f}{r + d + (1 - r - d)e_j f}. \tag{5}$$

Implied persuasion rate from the treatment-control gap:

$$f = \frac{v_T - v_C}{(e_T - e_C)(1-r)} \cdot \frac{(1-r)\, t_C\, t_T}{d}, \tag{6}$$

where turnout in town j is $t_j = r + d + (1 - r - d)e_j f$.

How I interpret (6):

- Numerator $v_T - v_C$ is the *reduced-form* vote-share effect.
- Denominator $e_T - e_C$ is the *first-stage* effect of cable carriage on Fox viewership.
- The extra factor adjusts for the fact that convincing a Democrat changes the two-party share differently than mobilizing a nonvoter.
- In short: persuasion $\approx$ (vote shift) / (extra exposure), with a turnout/composition correction.

3.6 (7) "First stage" for Fox News audience (Scarborough)

Town-level Fox audience regression:

$$e_k^{FOX} = \alpha + \delta_F\, d_{k,2000}^{FOX} + \Gamma_{2000} X_{k,2000} + \Gamma_{00-90} X_{k,00-90} + \Gamma_C C_{k,2000} + \varepsilon_k. \tag{7}$$

Interpret (7):

- δ_F here measures how much *cable availability* increases Fox News audience in that town.
- They estimate this with Scarborough "diary" audience and then scale to "recall" audience using a conversion factor.

3.7 (8) Generic persuasion formula used for turnout field experiments

If treatment exposure increases turnout from t_C to t_T with exposure rate $e_T - e_C$, then

$$f = \frac{t_T - t_C}{(e_T - e_C)\,(1 - t_C)}. \tag{8}$$

Interpret (8):

- This is the same persuasion logic: among people who would otherwise not do the action, what fraction are induced by treatment?

4 Figures and tables: what each one is saying (my interpretation)

4.1 Figure I: Fox News availability by county in 2000

Read of Figure I:

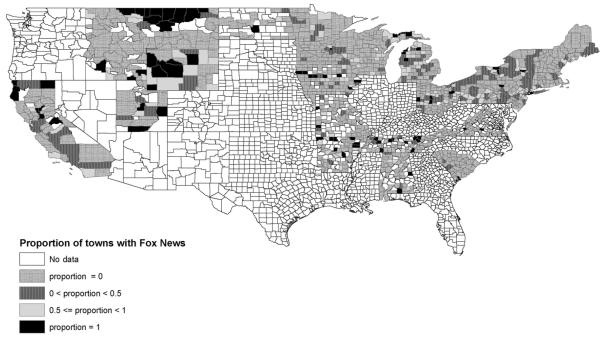

FIGURE I
Fox News Availability by County, 2000

TABLE I
SUMMARY STATISTICS ON FOX NEWS AUDIENCE (SCARBOROUGH DATA)

	Summary statistics					
	All survey respondents			Matched zip-coded subsample		
Sample	All (1)	Fox News recall audience (2)	Fox News non-audience (3)	All (4)	Fox News diary audience (5)	Fox News non-audience (6)
Cable variables						
Fox News (recall audience)	0.173	1	0	0.166*	0.591*	0.146*
	(0.379)			(0.372)	(0.493)	(0.353)
Fox News (diary audience)				0.035	1	0
				(0.185)		
CNN (recall audience)	0.341	0.619	0.283	0.353*	0.603*	0.341*
	(0.474)	(0.486)	(0.451)	(0.478)	(0.490)	(0.474)
CNN (diary audience)				0.103	0.350	0.094
				(0.304)	(0.478)	(0.292)
Demographic variables						
Some college	0.214	0.219	0.213	0.215	0.206	0.215
	(0.410)	(0.414)	(0.410)	(0.411)	(0.405)	(0.411)
College graduate	0.344	0.356	0.341	0.386	0.452	0.384
	(0.475)	(0.479)	(0.474)	(0.487)	(0.498)	(0.486)
African American	0.097	0.111	0.094	0.084	0.020	0.086
	(0.296)	(0.314)	(0.292)	(0.277)	(0.140)	(0.281)
Hispanic	0.107	0.081	0.112	0.180	0.094	0.183
	(0.309)	(0.273)	(0.315)	(0.384)	(0.293)	(0.387)
Unemployment	0.022	0.018	0.023	0.023	0.005	0.024
	(0.147)	(0.134)	(0.149)	(0.151)	(0.070)	(0.153)
Age	45.679	49.744	44.827	44.506	51.727	44.241
	(16.633)	(16.995)	(16.429)	(16.443)	(16.362)	(16.386)
Male	0.424	0.481	0.412	0.427	0.526	0.423
	(0.494)	(0.500)	(0.492)	(0.495)	(0.500)	(0.494)
Political variables						
Republican	0.282	0.375	0.262	0.267	0.536	0.257
	(0.450)	(0.484)	(0.440)	(0.442)	(0.499)	(0.437)
Democrat	0.319	0.294	0.324	0.335	0.159	0.342
	(0.466)	(0.455)	(0.468)	(0.472)	(0.366)	(0.474)
Voter turnout	0.693	0.769	0.677	0.677	0.819	0.672
	(0.461)	(0.421)	(0.468)	(0.468)	(0.386)	(0.470)
Subscriptions						
Cable	0.687	0.784	0.666	0.754	0.886	0.749
	(0.464)	(0.411)	(0.471)	(0.431)	(0.318)	(0.434)
Satellite	0.147	0.138	0.149	0.104	0.122	0.103
	(0.354)	(0.344)	(0.356)	(0.305)	(0.327)	(0.304)
No. observations	N = 105,201	N = 18,223	N = 86,968	N = 11,388	N = 403	N = 10,985

- Map shows, county-by-county, the fraction of towns with Fox News in their cable lineup in 2000.
- Visually, there is strong geographic clustering (whole regions with high vs low availability).
- Importantly, many counties are "mixed" (some towns have Fox, some do not). This within-county variation is what makes county fixed-effects designs feasible.

4.2 Table I: Summary statistics on Fox News audience (Scarborough data)

Interpretation of Table I:

- Two audience measures:
  - *Recall* audience: watched in past 7 days (inclusive, likely over-reports).
  - *Diary* audience: watched at least one 30-min block in diary week (stricter).
- Fox News audience size:

TABLE II
SUMMARY STATISTICS ON CABLE AND ELECTION DATA

	All sample			Mixed districts		Mixed counties	
	All towns (1)	Fox News in 2000 (2)	No Fox in 2000 (3)	Fox News in 2000 (4)	No Fox in 2000 (5)	Fox News in 2000 (6)	No Fox in 2000 (7)
Cable variables							
Number of channels	28.60	44.52	24.73	44.39	24.41	45.00	26.05
	(14.64)	(15.98)	(11.31)	(16.14)	(11.57)	(16.06)	(11.81)
Potential subscribers	78,124	163,622	57,384	140,457	47,373	167,006	70,832
	(149,015)	(246,661)	(103,131)	(198,871)	(91,025)	(254,926)	(116,337)
Voting variables							
Rep. vote share in 1996	0.470	0.479	0.467	0.482	0.475	0.477	0.475
	(0.125)	(0.125)	(0.125)	(0.124)	(0.124)	(0.125)	(0.127)
Rep. vote share in 2000	0.538	0.538	0.538	0.541	0.550	0.533	0.536
	(0.130)	(0.129)	(0.130)	(0.128)	(0.126)	(0.129)	(0.133)
Log turnout in 1996	−0.583	−0.583	−0.583	−0.583	−0.578	−0.573	−0.566
	(0.481)	(0.482)	(0.480)	(0.471)	(0.732)	(0.434)	(0.418)
Log turnout in 2000	−0.522	−0.525	−0.521	−0.525	−0.518	−0.510	−0.504
	(0.491)	(0.497)	(0.490)	(0.487)	(0.483)	(0.449)	(0.431)
Census variables for 2000							
Population	9,612	11,516	9,150	10,564	7,157	11,872	12,266
	(32,661)	(32,427)	(32,703)	(31,000)	(23,261)	(33,678)	(37,678)
Some college	0.257	0.259	0.257	0.258	0.257	0.258	0.254
	(0.064)	(0.063)	(0.064)	(0.064)	(0.066)	(0.063)	(0.067)
College	0.195	0.220	0.189	0.216	0.178	0.224	0.210
	(0.133)	(0.147)	(0.129)	(0.145)	(0.118)	(0.150)	(0.146)
African American	0.033	0.031	0.034	0.028	0.027	0.030	0.026
	(0.095)	(0.082)	(0.098)	(0.073)	(0.083)	(0.084)	(0.072)
Hispanic	0.031	0.035	0.030	0.032	0.027	0.035	0.041
	(0.073)	(0.072)	(0.073)	(0.067)	(0.065)	(0.074)	(0.096)
Unemployed	0.051	0.051	0.052	0.051	0.052	0.050	0.053
	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.036)	(0.038)
Urban	0.406	0.537	0.374	0.518	0.331	0.556	0.441
	(0.438)	(0.447)	(0.429)	(0.446)	(0.416)	(0.446)	(0.447)
Census variables, change from 1990 to 2000:							
Population	704	772	687	681	584	805	934
	(3,457)	(3,775)	(3,375)	(3,499)	(2,727)	(3,974)	(4,461)
Some college	0.040	0.035	0.041	0.036	0.044	0.035	0.034
	(0.048)	(0.046)	(0.049)	(0.046)	(0.049)	(0.046)	(0.047)
College	0.037	0.041	0.036	0.041	0.035	0.042	0.041
	(0.042)	(0.044)	(0.042)	(0.045)	(0.041)	(0.045)	(0.043)
African American	0.004	0.003	0.004	0.003	0.003	0.003	0.004
	(0.026)	(0.025)	(0.026)	(0.024)	(0.024)	(0.024)	(0.024)
Hispanic	0.011	0.013	0.010	0.012	0.010	0.013	0.011
	(0.026)	(0.030)	(0.025)	(0.027)	(0.025)	(0.029)	(0.028)
Unemployed	−0.012	−0.011	−0.013	−0.011	−0.013	−0.012	−0.012
	(0.038)	(0.037)	(0.039)	(0.037)	(0.040)	(0.038)	(0.036)
Urban	0.082	0.079	0.083	0.082	0.080	0.084	0.086
	(0.238)	(0.239)	(0.238)	(0.241)	(0.240)	(0.247)	(0.242)
No. of observations	$N = 9{,}256$	$N = 1{,}807$	$N = 7{,}449$	$N = 1{,}734$	$N = 5{,}897$	$N = 1{,}548$	$N = 2{,}342$

  - Recall: Fox $\approx 17.3\%$ vs CNN $\approx 34.1\%$ (Fox already about half of CNN by 2000).
  - Diary (in matched subsample): Fox $\approx 3.5\%$ vs CNN $\approx 10.3\%$ (diary is 3–5$\times$ smaller than recall).

- Fox viewers skew older and more male.
- Politically, Fox viewers are substantially more Republican (both recall and diary), and report higher turnout.
- Cable/satellite: Fox viewers are more likely to subscribe to cable; satellite rates are similar-ish.
- Big caveat: the Republican skew among Fox viewers could be (i) selection (Republicans choose Fox) and/or (ii) persuasion (Fox changes party ID). The paper cannot fully separate these with this table alone.

4.3 Table II: Summary statistics on cable and election data (town level)

Interpretation of Table II:

- This table documents that Fox towns differ from non-Fox towns in important observables:
  - Fox towns have *many more channels* (about 44.5 vs 24.7).
  - Fox towns have much larger cable systems (potential subscribers).
  - Fox towns are more urban and somewhat larger in population.
- Raw voting trends: unconditionally, Fox towns do *not* look like they got more Republican than non-Fox towns between 1996 and 2000. This is a warning that selection/confounding is real and controls are needed.

TABLE III
DETERMINANTS OF FOX NEWS AVAILABILITY, LINEAR PROBABILITY MODEL

	Availability of Fox News via cable in 2000						
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Pres. republican vote share in 1996	0.1436	0.6363	0.3902	−0.0343	−0.0442	0.0902	0.0627
	(0.1549)	(0.2101)***	(0.1566)**	(0.0937)	(0.1024)	(0.1321)	(0.1333)
Pres. log turnout in 1996	0.1101	0.0909	0.0656	0.0139	−0.0053	0.0286	0.0257
	(0.0557)**	(0.0348)***	(0.0278)**	(0.0124)	(0.0173)	(0.0234)	(0.0258)
Pres. Rep. vote share change 1996–1992						0.214	−0.2548
						(0.2481)	(0.2345)
Control variables							
Census controls: 1990 and 2000	—	X	X	X	X	X	X
Cable system controls	—	—	X	X	X	X	X
U. S. House district fixed effects	—	—	—	X	—	X	—
County fixed effects	—	—	—	—	X	—	X
F-test: Census controls = 0		$F = 3.54$***	$F = 2.73$***	$F = 1.11$	$F = 1.28$	$F = 1.57$**	$F = 1.31$
F-test: Cable controls = 0			$F = 18.08$***	$F = 21.09$***	$F = 18.61$***	$F = 8.19$***	$F = 8.75$***
R^2	0.0281	0.0902	0.4093	0.6698	0.7683	0.6313	0.7622
N	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 3{,}722$	$N = 3{,}722$

TABLE IV
THE EFFECT OF FOX NEWS ON THE 2000–1996 PRESIDENTIAL VOTE SHARE CHANGE

	Republican two-party vote share change between 2000 and 1996 pres. elections						
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Availability of Fox News via cable in 2000	−0.0025	0.0027	0.008	0.0042	0.0069	0.0037	0.0048
	(0.0037)	(0.0024)	(0.0026)***	(0.0015)***	(0.0014)***	(0.0021)*	(0.0019)**
Pres. Rep. vote share change 1988–1992						0.0229	0.0514
						(0.0216)	(0.0219)**
Constant	0.0347	−0.028	−0.0255	0.0116	0.0253	−0.0377	0.0081
	(0.0017)***	(0.0245)	(0.0236)	(0.0154)	(0.0185)	(0.0258)	(0.0313)
Control variables							
Census controls: 1990 and 2000	—	X	X	X	X	X	X
Cable system controls	—	—	X	X	X	X	X
U. S. House district fixed effects	—	—	—	X	—	X	—
County fixed effects	—	—	—	—	X	—	X
R^2	0.0007	0.5207	0.5573	0.7533	0.8119	0.7528	0.8244
N	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 9{,}256$	$N = 3{,}722$	$N = 3{,}722$

- "Mixed districts" / "mixed counties" columns explain where identification comes from:
  - District-FE design uses within-district variation (many districts are mixed).
  - County-FE design uses within-county variation (fewer counties are mixed ⇒ smaller sample but tighter geographic comparison).

4.4 Table III: Determinants of Fox News availability (selection test)

Interpretation of Table III:

- Columns (1)–(3): without fine geographic FE, Fox availability correlates with pre-period politics and turnout (especially after adding demographics).
- Key point: once they add cable controls and district FE (col. (4)) or county FE (col. (5)), the coefficients on:
  - 1996 Republican vote share, and
  - 1996 turnout

  become small and statistically insignificant (even slightly negative).
- So the story is: Fox selected into *bigger cable systems* and into certain areas, but *within* an area, conditional on system size, the remaining rollout is close to idiosyncratic.
- Placebo trend check (cols. (6)–(7)): earlier voting trend (1988–1992) does not predict Fox availability either.

4.5 Table IV: Main result — Fox News and 2000–1996 presidential vote-share change

Interpretation of Table IV:

TABLE V
THE FOX NEWS EFFECT: ROBUSTNESS AND PERSISTENCE

		Robustness				
		Pres. Rep. vote share change 2000–1996				
	Rep. two-party vote share in 2000	All-party vote share	Two-party vote share			Persistence Pres. Rep. vote share 2004–2000
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)
Availability of Fox News via cable in 2000	0.0041	0.004	0.0048	0.0041	0.0047	0.0021
	(0.0016)***	(0.0016)**	(0.0016)***	(0.0017)***	(0.0016)***	(0.0020)
Republican vote share in 1996	0.9362					
	(0.0079)***					
Control variables						
Census controls: 1990 and 2000	X	X	X	X	X	X
Cable system controls	X	X	X	X	X	X
U. S. House district fixed effects	X	X	X	X	X	X
Election data with high coverage	—	—	X	—	—	—
Unweighted, turnout > 2000	—	—	—	X	—	—
Nearest-neighbor matching, unweighted	—	—	—	—	X	—
R^2	0.9824	0.827	0.7556	0.7369	—	0.6281
N	N = 9,256	N = 9,256	N = 7,758	N = 3,241	N = 9,256	N = 8,605

- Column (1) (no controls) even shows a slightly negative association (Fox towns trend a bit less Republican), consistent with confounding.
- Once cable controls are included (col. (3)) and especially with geographic FE:
  - District FE (col. (4)): $\hat{\delta}_F \approx 0.0042$ (about +0.42pp), highly significant.
  - County FE (col. (5)): $\hat{\delta}_F \approx 0.0069$ (about +0.69pp), highly significant.
- Interpretation: *cable availability of Fox* increases the Republican two-party presidential vote share by roughly 0.4–0.7 percentage points in 2000 relative to 1996.
- Adding very old trends (1988–1992) (cols. (6)–(7)) reduces precision (smaller N), but the Fox coefficient remains positive and in the same ballpark.

4.6 Table V: Robustness and persistence

Interpretation of Table V:

- Robustness to outcome definition:
  - Using level outcome in 2000 controlling for 1996 (col. (1)) still gives a Fox effect ≈ 0.0041.
  - Using all-party vote share change (col. (2)) gives a similar effect ≈ 0.004.
- Robustness to sample/estimator:
  - Restricting to states with higher election-data coverage (col. (3)): effect slightly larger.
  - Dropping very small towns (unweighted, turnout ≥ 2000) (col. (4)): effect similar.
  - Nearest-neighbor matching (col. (5)): effect similar to OLS benchmark.
- Persistence (col. (6)): Fox availability in 2000 is associated with a small positive (but statistically insignificant) additional Republican shift from 2000 to 2004, suggesting effects do not obviously fade out quickly.

4.7 Table VI: Interactions (heterogeneity) and placebo tests

Interpretation of Table VI:

- Heterogeneity:

TABLE VI
THE FOX NEWS EFFECT: INTERACTIONS AND PLACEBO SPECIFICATIONS

	Interactions		Placebo specifications		
	Presid. Rep. vote share 2000–1996		Presidential Republican vote share		
			2000–1996	1996–1992	1992–1988
Dep. var.	(1)	(2)	(3)	(4)	(5)
Availability of Fox News via cable in 2000	0.0109	0.0105	0.0036	−0.0024	0.0026
	(0.0042)***	(0.0039)***	(0.0016)**	(0.0031)	(0.0026)
Availability of Fox News via cable in 2003			−0.0001		
			(0.0012)		
Fox News in 2000* (no. of channels/10)	−0.0019	−0.0013			
	(0.0008)**	(0.0008)			
Fox News in 2000* (urban in 2000)	0.004	0.0033			
	(0.0023)*	(0.0020)*			
Fox News in 2000* (swing district)	−0.0043	−0.0004			
	(0.0030)	(0.0023)			
Fox News in 2000* (Republican district)	−0.0028	−0.0064			
	(0.0027)	(0.0024)***			
Control variables:					
Census controls: 1990 and 2000	X	X	X	X	X
Cable system controls	X	X	X	X	X
U. S. House district fixed effects	X	—	X	X	X
County fixed effects	—	X	—	—	—
R^2	0.754	0.8124	0.7523	0.6197	0.6517
N	$N = 9{,}256$	$N = 9{,}256$	$N = 8{,}645$	$N = 4{,}006$	$N = 3{,}722$

  - # channels interaction is negative: with more cable channels (more "competition"/choice), the Fox effect is smaller.
  - Urban interaction is positive: effect is larger in more urban places.
  - Political environment: relative to Democratic districts, the Fox effect is smaller in more Republican districts (negative interaction; strongest with county FE).

- Placebos:
  - Fox availability in 2003 does not predict the 1996–2000 change once control for Fox in 2000 (good placebo).
  - Fox availability in 2000 does *not* predict vote-share changes in pre-Fox periods (1992–1996, 1988–1992).

- Overall: supports the parallel-trends / no-pretrends story.

4.8 Table VII: Turnout and Senate elections

Interpretation of Table VII: Turnout (cols. (1)–(3))

- District FE estimate is small and not significant (col. (1)).
- County FE estimate is larger and significant (col. (2)), suggesting a turnout increase on the order of 1–2% in votes cast.
- Heterogeneity by district partisanship (col. (3)):
  - Base effect is for Democratic districts.
  - Negative interactions for swing/Republican districts imply the turnout effect is *strongest in Democratic districts.*

Interpretation of Table VII: Senate vote share (cols. (4)–(6))

- Fox News increases Republican *Senate* vote share by about 0.7–1.0 percentage points (depending on FE), and it is statistically significant.
- The New York featured race interaction is small and not significant: no clear evidence that Fox's heavy candidate-specific coverage generates an extra bump beyond the general effect.
- Negative interactions for swing/Republican districts imply the Senate effect is strongest in Democratic districts (consistent with an "at-risk" non-Republican audience being larger there).

TABLE VII
FOX NEWS AND OTHER POLITICAL OUTCOMES: TURNOUT AND SENATORIAL ELECTIONS

	Turnout (presidential elections)			Senatorial elections		
	Change in log (total votes cast) between 2000 and 1996 pres. elections			Republican vote share in 2000 senatorial elections		
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)
Availability of Fox News via cable in 2000	0.0046	0.0178	0.0147	0.0072	0.0071	0.01
	(0.0039)	(0.0051)***	(0.0061)**	(0.0026)***	(0.0028)**	(0.0035)***
Change in log (voting-age population) bw. 1996 and 2000	0.3655	0.3707	0.3641			
	(0.0427)***	(0.0440)***	(0.0425)***			
Fox News in 2000* (New York race)				0.0039	−0.0017	0.0033
				(0.0067)	(0.0060)	(0.0067)
Republican vote share in 1996 presidential elections				0.8295	0.8432	0.8289
				(0.0111)***	(0.0146)***	(0.1111)***
Fox News in 2000* (swing district)			−0.0207			−0.0042
			(0.0087)*			(0.0047)
Fox News in 2000* (Republican district)			−0.0177			−0.0075
			(0.0090)**			(0.0054)
Control variables						
Census controls: 1990 and 2000	X	X	X	X	X	X
Cable system controls	X	X	X	X	X	X
U. S. House district fixed effects	X	—	X	X	—	X
County fixed effects	—	X	—	—	X	—
R^2	0.6151	0.6863	0.658	0.9768	0.9829	0.9768
N	N = 9,256	N = 9,256	N = 9,256	N = 8,192	N = 8,192	N = 8,192

TABLE VIII
THE EFFECT OF FOX NEWS EXPOSURE ON FOX NEWS AUDIENCE: PERSUASION RATES (SCARBOROUGH DATA)

	Share of town respondents that watched at least thirty minutes of a channel in past week						
	Watched Fox News				Watched CNN		
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Availability of Fox News via cable in 2000	0.027	0.0371	0.0251	0.0346	0.0251	0.0042	0.0045
	(0.0058)***	(0.0105)***	(0.0082)***	(0.0116)***	(0.0121)**	(0.0114)	(0.0104)
Availability of Fox News via cable in 2003				0.0016			
				(0.0090)			
Constant	0.0262	0.0928	−0.0531	0.1317	0.0947	0.7213	0.0917
	(0.0036)***	(0.1046)	(0.1316)	(0.1102)	(0.0054)***	(0.1674)***	(0.2119)
Control variables							
Census controls: 1990 and 2000	—	X	X	X	—	X	X
Cable system controls	—	X	X	X	—	X	X
U. S. House district fixed effects	—	X	—	X	—	X	—
County fixed effects	—	—	X	—	—	—	X
2SLS estimates of persuasion rates							
Estimates using (predicted) recall audience measure		0.0339	0.0827				
		(0.0151)**	(0.0298)***				
Estimates using diary audience measure		0.1162	0.2829				
		(0.0518)**	(0.1020)***				
R^2	0.0655	0.3105	0.3507	0.3148	0.0217	0.3872	0.4262
N	N = 568	N = 568	N = 568	N = 545	N = 568	N = 568	N = 568

4.9 Table VIII: Effect of Fox carriage on Fox audience + implied persuasion rates

Interpretation of Table VIII: first stage (audience)

- When Fox is carried on cable, the Fox *diary* audience share rises by about 2.5–3.7 percentage points, depending on FE.
- There is substantial non-cable viewing too (the constant in non-Fox cable towns is positive), consistent with satellite access or measurement noise.
- Placebos/validity checks:
  - "Fox in 2003" does not predict Fox audience in 2000 (placebo).
  - Fox availability does not raise CNN audience once controls are included (good "no effect on other channels" check).

Interpretation of Table VIII: persuasion rates (2SLS-style ratio)

- They combine:
  - reduced form: Fox effect on voting (Table IV), and
  - first stage: Fox effect on audience (Table VIII),

  plus political composition adjustments from the persuasion model.
- Using a *predicted recall* audience measure (diary scaled up), persuasion is about:
  - $f \approx 0.034$ with district FE,
  - $f \approx 0.083$ with county FE.

TABLE IX
COMPARISON WITH PERSUASION RATES IN OTHER MEDIA STUDIES

	Variable: persuasion rate f (share of listeners convinced by media)									
Paper	Treatment (1)	Election type or question (2)	Variable t (3)	Year (4)	Place (5)	Sample size (6)	Control group t_T (7)	Treatment group t_C (8)	Exposure rate $e_T - e_C$ (9)	Persuasion rate (10)
Fox News Study										
DellaVigna and Kaplan [2006]	Fox News exposure, district f.e.					N = 66,372,804	0.556	0.560	0.127	0.034/0.116
	Fox News exposure, county f.e.	Presidential election	Republican vote share	2000	28 U. S. states		0.556	0.563	0.086	0.083/0.283
Field Experiments										
Gerber and Green [2000]	Door-to-door canvassing	Federal elect.	Turnout	1998	New Haven	N = 14,473	0.422	0.463	0.270	0.263
	Canvassing + mail + calls	Federal elect.	Turnout	1998	New Haven	N = 14,850	0.422	0.448	0.270	0.167
Green, Gerber, and Nickerson [2003]	Door-to-door canvassing	Local elect.	Turnout	2001	6 cities	N = 18,933	0.286	0.310	0.293	0.118
Green and Gerber [2001]	Phone calls by youth vote	General elect.	Turnout	2000	4 cities	N = 4,377	0.660	0.711	0.737	0.205
	Phone calls 18–30-year-olds	General elect.	Turnout	2000	2 cities	N = 4,377	0.405	0.416	0.414	0.045
Gerber, Karlan, and Bergan [2006]	Free subscription to *Washington Post*	Governor elect.	Dem. share of votes	2005	Washington	N = 1,011	0.291	0.363	0.940	0.109
Laboratory Experiments										
Ansolabehere and Iyengar [1995]	Laboratory exposure to 30-second political ad	Governor elect. Senate elect. Mayor elect.	Vote share for party sponsoring ad	1990 1992 1993	Southern California	N=1,716	0.530	0.568	1.000	0.082
Surveys										
Kull, Ramsay, and Lewis [2003]	Respondent watches Fox News	Did U. S. find WMD in Iraq?	Share of yes answers	2003	USA	N=8,634	0.220	0.330	1.000	0.141
Gentzkow and Shapiro [2004]	Respondent watches CNN	Did Arabs do 9/11 attack?	Share of yes answers	2002	Arab countries	N=2,457	0.215	0.280	1.000	0.083
	Respondent watches Al Jazeera			2002		N=2,457	0.215	0.133	1.000	0.105

- Using the stricter *diary* measure, persuasion is larger:
  - $f \approx 0.116$ (district FE),
  - $f \approx 0.283$ (county FE).
- Read this as: among non-Republicans who are exposed, somewhere between a few percent and (with stricter exposure definitions) a few-tenths are persuaded to vote Republican.

4.10 Table IX: Comparing persuasion rates in other studies

Interpretation of Table IX:

- They compute comparable persuasion rates across:
  - voter-mobilization field experiments (canvassing/phone calls),
  - newspaper-subscription field experiment,
  - lab exposure to political ads,
  - surveys relating media consumption to beliefs.
- The big picture: Fox persuasion estimates (roughly 0.03–0.08 using recall; 0.11–0.28 using diary) sit in the same order of magnitude as many experimental and survey-based persuasion rates.
- Caveat they acknowledge: survey-based persuasion could be confounded by selection (sorting), but the magnitudes line up with experiments anyway.

4.11 Appendix II: Full list of controls and coefficients (for Tables III and IV benchmark specs)

Interpretation of Appendix II:

APPENDIX II: List of Controls for Tables III and IV

	Regression of Fox News on determinants (Table III)		Regression of vote share on Fox News (Table IV)	
	Coefficient	Std. error	Coefficient	Std. error
	(1)	(2)	(3)	(4)
Census controls 2000				
Population	−0.0004	(0.0013)	−0.0003	(0.0001)***
High school	0.0334	(0.2233)	0.0252	(0.0160)
Some college	−0.0218	(0.1964)	−0.0664	(0.0141)***
College graduate	0.0672	(0.1599)	−0.097	(0.0103)***
Male	0.2285	(0.2859)	0.1146	(0.0208)***
African American	−0.013	(0.1086)	−0.0547	(0.0061)***
Hispanic	−0.1155	(0.1973)	−0.0571	(0.0130)***
Employment	0.0759	(0.1225)	0.0367	(0.0089)***
Unemployment	0.3511	(0.3768)	0.1212	(0.0265)***
Married	−0.1057	(0.1419)	0.0033	(0.0078)
Income	−0.0124	(0.0072)*	−0.0012	(0.0004)***
Urban	−0.0293	(0.0251)	−0.0073	(0.0014)***
Census controls 2000–1990				
Population	−0.0189	(0.0106)*	0.0018	(0.0005)***
High school	0.0132	(0.1997)	0.0136	(0.0136)
Some college	−0.1716	(0.1931)	0.0596	(0.0154)***
College graduate	0.0351	(0.2097)	0.0841	(0.0127)***
Male	−0.218	(0.3099)	−0.0311	(0.0238)
African American	−0.3921	(0.2173)*	0.0687	(0.0181)***
Hispanic	−0.0138	(0.3556)	−0.054	(0.0206)***
Employment	−0.0367	(0.1453)	−0.0209	(0.0109)*
Unemployment	−0.3466	(0.2663)	−0.0594	(0.0229)***
Married	−0.0786	(0.1238)	−0.0105	(0.0111)
Income	0.0157	(0.0118)	0.0011	(0.0010)
Urban	−0.0039	(0.0280)	0.0029	(0.0018)
Decile in potential subscribers				
Decile 2	−0.006	(0.0188)	−0.0007	(0.0032)
Decile 3	0.0534	(0.0249)**	−0.0006	(0.0031)
Decile 4	0.0499	(0.0295)*	−0.0022	(0.0031)
Decile 5	0.0388	(0.0334)	−0.0053	(0.0033)
Decile 6	0.0563	(0.0369)	−0.006	(0.0033)*
Decile 7	0.0259	(0.0410)	−0.011	(0.0034)***
Decile 8	0.022	(0.0387)	−0.0105	(0.0035)***
Decile 9	−0.0056	(0.0432)	−0.0124	(0.0035)***
Decile 10	0.1786	(0.0481)***	−0.0127	(0.0035)***
Decile in number of channels				
Decile 2	0.0355	(0.0221)	0.0024	(0.0019)
Decile 3	0.0778	(0.0267)***	0.002	(0.0021)
Decile 4	0.051	(0.0287)*	−0.0005	(0.0023)
Decile 5	0.0394	(0.0344)	−0.0011	(0.0021)
Decile 6	0.1122	(0.0278)***	0.0001	(0.0019)
Decile 7	0.3347	(0.0464)***	0.0015	(0.0022)
Decile 8	0.4292	(0.0526)***	−0.0017	(0.0023)
Decile 9	0.4113	(0.0541)***	−0.0025	(0.0023)
Decile 10	0.6475	(0.0500)***	−0.0014	(0.0025)
U. S. House district fixed effects	X	X	X	X

APPENDIX III: The Fox News Effect: Comparison with Earlier Results

	Presidential republication vote share change between 2000 and 1996					
Dep. var.	(1)	(2)	(3)	(4)	(5)	(6)
Availability of Fox News via cable in 2000	0.0014	0.004	0.0002	0.0022	−0.0003	0.0017
	(0.0016)	(0.0015)***	(0.0015)	(0.0014)	(0.0016)	(0.0014)
Control variables						
Census controls: 1990 and 2000	X	X	X	X	X	X
Cable system controls	X	X	X	X	X	X
U. S. House district fixed effects	X	—	X	—	X	—
County fixed effects	—	X	—	X	—	X
Specifications						
Unweighted	X	X	X	X	X	X
Include high-measurement-error obs.			X	X	X	X
Exclude HI, ND, NJ, WY					X	X
R^2	0.5666	0.6796	0.5574	0.6765	0.5371	0.6641
N	N = 9,256	N = 9,256	N = 9,802	N = 9,802	N = 9,131	N = 9,131

- This is mainly "plumbing":
  - which exact Census and cable controls enter,
  - and what their coefficients look like in the benchmark district-FE specifications.
- The biggest predictors of Fox carriage are the cable-system deciles (especially high deciles of # channels), which matches the idea that capacity constraints and system size drive rollout.
- Many demographics correlate with Republican vote share changes, which is why controlling for them matters for precision and to avoid omitted-variable bias.

4.12 Appendix III: Why earlier drafts found smaller/null effects

Interpretation of Appendix III:

- Three things push estimates toward zero in earlier drafts:
  1. Unweighted regressions (small rural towns get a lot of weight and appear to have smaller effects / more noise).
  2. Including towns with high measurement error in Fox availability and/or voting data.
  3. A slightly different state sample.
- Once they weight by turnout and drop high-error observations, the Fox effect becomes clearer and statistically significant in their preferred specs.

5 Takeaway

- Identification: compare changes 1996→2000 across towns that do vs. do not get Fox by 2000, controlling heavily for cable-system size + demographics + area FE (district/county).
- Main reduced form: Fox carriage ⇒ +0.4 to +0.7 pp Republican two-party vote share in the presidential election.
- Mechanisms:
  - Some evidence of increased turnout (especially in Democratic districts).
  - Senate effects suggest a broader ideological shift, not only candidate-specific persuasion.
- Persuasion magnitude: depending on how exposure is defined, Fox convinces on the order of a few percent up to a few-tenths of its *non-Republican* viewers to vote Republican.

6 Conclusion and summary

6.1 What the paper concludes (in plain language)

The main takeaway is that **media bias can causally move elections**, even in a mature media environment. Using the staggered rollout of *Fox News Channel* across local cable systems from 1996 to 2000, the paper finds that **towns that got Fox News by 2000 shifted toward Republicans** relative to similar towns that did not.

Concretely:

- **Presidential vote share:** Fox News availability raises the Republican two-party vote share change (2000–1996) by about **0.4 to 0.7 percentage points** (depending on whether identification is within House districts or within counties).
- **Senate vote share:** Fox News also increases the Republican vote share in Senate races (including races not heavily covered nationally), suggesting the effect is not just candidate-specific, but more like a **general ideological shift**.
- **Turnout:** there is evidence that Fox News **increased turnout**, especially in more Democratic districts (mobilization channel), although turnout results are not as uniformly tight as the vote-share results.
- **Persuasion magnitudes:** combining the reduced-form vote shift with audience data, the implied persuasion rate is sizeable:
  - **3–8%** of *non-Republican* viewers persuaded (using a more inclusive "recall" audience concept), or
  - **11–28%** using the stricter "diary" audience measure.

Overall, the conclusion is that **biased political information can measurably shift voting behavior** in the aggregate, and the implied individual-level persuasion effects are large enough to matter in close elections.

6.2 Final summary of the paper (what it does + what it finds)

Research question. Does biased media affect political behavior, or do rational voters fully "filter out" slant?

Setting and data. Fox News launched in late 1996 and expanded quickly. Because cable is locally organized and capacity-constrained, **neighboring towns often differed in whether Fox was carried by 2000**. The authors build a large town-level panel (9,256 towns across 28 states), matching: (i) cable lineups (Nov. 2000), (ii) election outcomes (especially 1996 and 2000), and (iii) Census demographics (levels and changes).

Empirical strategy. They use a **difference-in-differences design** at the town level: compare the change in Republican vote share from 1996 to 2000 between towns with Fox by 2000 and towns without Fox by 2000, controlling for:

- cable-system size/capacity (deciles of channels and subscribers),
- demographics (levels and 1990–2000 changes), and
- **geographic fixed effects** (House district FE, and even tighter county FE).

They also run selection tests and placebo tests to check pre-trends.

Core findings. After controls and geographic FE:

- Fox News availability has a **positive and statistically strong** effect on the Republican presidential vote share in 2000 relative to 1996.
- The effect is **larger in more Democratic areas** (consistent with "more persuadable/mobilizable" voters there) and **smaller where there are more channels** (consistent with competition/choice diluting the message).
- Senate results show the effect is **not only about a single candidate** heavily covered on Fox; instead it looks like a broader ideological impact.

Interpretation. Using audience data, they translate the reduced-form vote shift into a persuasion rate: a nontrivial fraction of viewers who were not already Republican voters appear to switch (or be mobilized to vote Republican).

6.3 Intuition (why their design is supposed to work)

The intuition is basically: **Fox News enters some cable systems earlier than others for reasons related to the cable market (capacity, deals, timing), not because those particular towns were already on a unique Republican trend**.

So the strategy is credible if two things are true:

1. **Conditional quasi-random rollout:** After controlling for cable-system characteristics and local geography (district/county), *Fox availability is not predicted by prior political outcomes or trends.*
2. **Parallel trends:** In the absence of Fox, treated and control towns within the same area would have had similar changes in vote share.

They support these with:

- **Selection regressions:** once I control for cable size and geographic FE, 1996 vote share and turnout do not meaningfully predict Fox availability.
- **Placebos:** Fox availability in 2000 does not "predict" vote changes in pre-Fox periods (e.g., 1992–1996, 1988–1992), and Fox availability in 2003 does not explain the 1996–2000 shift once 2000 availability is controlled for.

6.4 What mechanism is most consistent with the evidence?

The paper cannot perfectly decompose mechanisms, but the pattern points to two channels:

- **Persuasion / switching:** some Democratic (or independent) voters exposed to Fox move toward Republican candidates.
- **Mobilization:** turnout increases in some specifications, and the turnout effect is stronger in Democratic districts, consistent with Fox activating conservative-leaning nonvoters who live in otherwise Democratic areas.

The Senate evidence reinforces the idea that this is not only "candidate-specific advertising," but a **general shift in beliefs/ideology**.

6.5 How the authors interpret the persuasion effect (rational vs. behavioral)

They present (at least) two broad interpretations:

1. **Rational learning:** if voters are initially uncertain about how biased Fox is, they may partially treat pro-Republican coverage as informative about candidate quality, causing a *temporary* shift that should fade as voters learn Fox's slant.
2. **Nonrational persuasion:** voters do not sufficiently discount biased information, so exposure causes a *more persistent* shift in beliefs and voting.

They argue that the **persistence evidence** (Fox exposure in 2000 is associated with continued Republican shifts into 2004, though not always precisely estimated) fits better with persuasion than with a pure learning story, but they do not claim to decisively separate the two.

6.6 What the paper leaves open

The conclusion explicitly flags open questions that I should remember:

- They focus on the **extensive margin** (votes/turnout), not on changes in intensity of political commitment among existing partisans (e.g., donations).
- They do not directly measure **policy outcomes** from the electoral shift.
- It would be valuable to study similar effects from other biased media sources (local newspapers, talk radio, etc.) to understand generality.